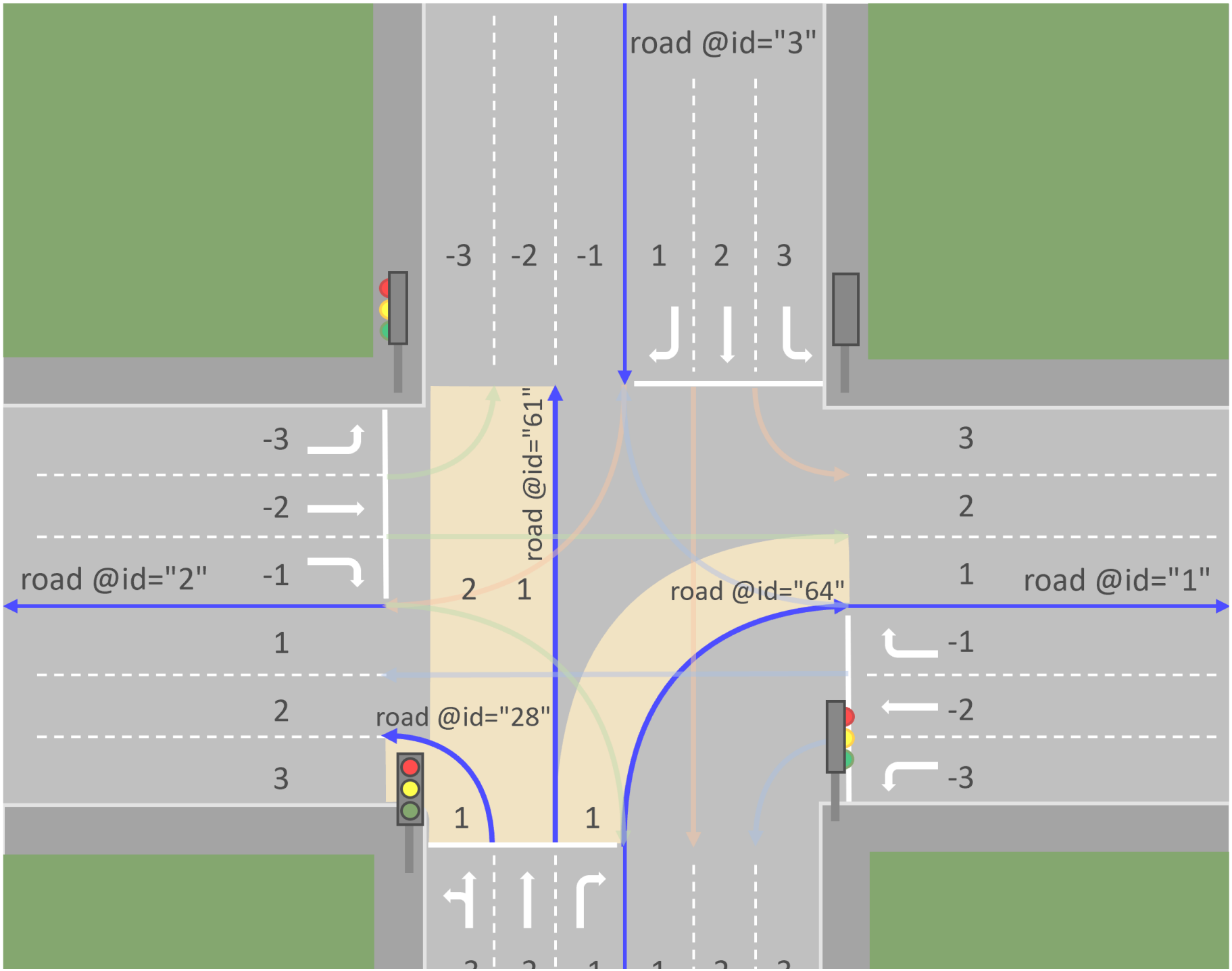
OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

12.4 Connecting roads

Connecting roads link the roads that meet in a junction. They describe the paths that a vehicle can travel across a junction. Connecting roads are modeled in the same way as standard roads.

The paths described by a connecting road is based on its lanes. The connecting road specifies the connections between the lanes of an incoming road and the lanes of an outgoing road of the same junction. If the lanes of an incoming and outgoing road are not linked, this means that there is no traversable path between these lanes.


road @id="3"
-3
-2
-1
1
2
3
road @id="61"
-3
-2
-1
road @id="2"
2
1
road @id="64"
3
2
1
road @id="1"
1
2
3
-1
-2
-3
road @id="28"
1
1

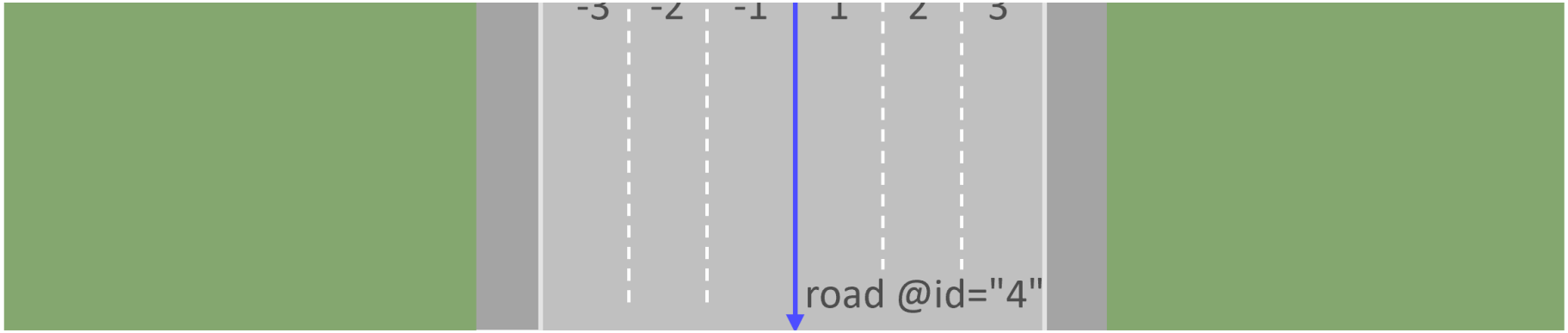


Figure 84. Connecting roads of junction with id 1 (left hand traffic)

Figure 83 and Figure 84 show the connecting roads inside the junction area that connect the incoming and outgoing roads.

NOTE

The example in Table 58, Table 59, and Table 60 only considers how to cross the junction from the road with id="4".

Table 58. Junction with id 1

Connection id	Incoming road	Connecting road	Contact point	Lane link from	Lane link to
9	4	28	start		
				-3	1
10	4	61	start		
				-2	1
				-3	2
11	4	64	start		
				-1	1

Table 59. Roads

Road id	Predecessor	Contact predecessor	Successor	Contact successor	Junction
1	junction with id 1				-1
2	junction with id 1				-1
3			junction with id 1		-1
4	junction with id 1				-1
28	road with id 4	start	road with id 2	start	1
61	road with id 4	start	road with id 3	end	1
64	road with id 4	start	road with id 1	start	1

Table 60. Lane links

Road id	Lane id	Predecessor's lane id	Successor's lane id
28	1	-3	3
61	1	-2	-2
61	2	-3	-3
64	1	-1	1
4	-3	no lane link	
4	-2	no lane link	
4	-1	no lane link	
1	1	no lane link	
3	-2		no lane link

Road id	Lane id	Predecessor's lane id	Successor's lane id
3	-3		no lane link
2	3	no lane link	

Elements in UML model

`<connection connectingRoad="…">` element

In ASAM OpenDRIVE, connecting roads are represented by the @connectingRoad attribute of `<connection>` elements within the `<junction>` element.

```
UML class: t_junction_connection_common
XML tag:   <connection> (Multiplicity: 0..*)
```

Provides information about a single connection within a common junction.

Table 61. Attributes of the <connection> element

Name	Type	Use	Description
`connectingRoad`	string	required	ID of the connecting road. Only to be used for junctions of @type="default".
`contactPoint`	e_contactPoint	optional	Contact point on the @connectingRoad or @linkedRoad. Required for all junction types except virtual.
`id`	string	required	Unique ID within the junction
`incomingRoad`	string	optional	ID of the incoming road. Required for all junction types except virtual.

`<laneLink>` element

In ASAM OpenDRIVE, lane links are represented by `<laneLink>` elements within the `<connection>` element.

```
UML class: t_junction_connection_laneLink
XML tag:   <laneLink> (Multiplicity: 0..*)
```

Provides information about the lanes that are linked between an incoming road and a connecting road. It is strongly recommended to provide this element. It is deprecated to omit the `<laneLink>` element.

Table 62. Attributes of the <laneLink> element

Name	Type	Use	Unit	Introduced	Description
`from`	integer	required			ID of the incoming lane
`overlapZone`	t_grZero	optional	m	1.8.0	Specifies the length of the area where traffic from both overlapping lanes shares the space. It is defined in s length relative to the position of the junction. Intended for direct junctions only. Default is 100.
`to`	integer	required			ID of the connection lane

XML example

- Ex_LHT-Complex-X-Junction.xodr (left-hand traffic)
- UC_Simple-X-Junction.xodr (right-hand traffic)

Rules

The following rules apply to connecting roads:

- asam.net:xodr:1.8.0:junctions.connection.one_link_to_incoming: Each connecting road shall be associated with at most one `<connection>` element per incoming road. A connecting road shall only have the `<laneLink>` element for that direction.
- A connecting road may have both right and left lanes.
- An incoming road with multiple lanes may be connected to the lanes of the road leading out off the junction in different ways:

- asam.net:xodr:1.7.0:junctions.connection.no_lane_change_for_mult_con_roads: By multiple connecting roads, each with one `<laneLink>` element for the connection between two specific lanes. Lane changes within this junction are not possible.
- asam.net:xodr:1.7.0:junctions.connection.lane_change_one_con_road: By one connecting road with multiple `<laneLink>` elements for the connections between the lanes.
- The linked lanes shall fit smoothly as described for roads (see Section 10.3, "Road linkage").
- The @connectingRoad attribute shall not be used for junctions with @type="direct".

Related topics

- Section 12.3, "Incoming roads"
- Section 12.1, "Introduction to junctions"
- Section 10.3, "Road linkage"
- Section 11.5, "Lane linkage"
- Section 12.4.1, “Priorities of connecting roads within a junction”
- Section 11.7.1, "Lane access"

12.4.1 Priorities of connecting roads within a junction

The `<priority>` element within the `<junction>` element defines the priority of a road over another road as a pair with the ID of the road with higher priority in the @high attribute and the ID of the road with lower priority in the @low attribute.

Elements in UML model

`<priority>` element

In ASAM OpenDRIVE, the priority of roads is represented by `<priority>` elements within the `<junction>` element.

```
UML class: t_junction_priority
XML tag:   <priority> (Multiplicity: 0..*)
```

The junction priority record provides information about the priority of one road over another road that are part of this junction. It is only required if priorities cannot be derived from signs or signals in a junction or on tracks leading to a junction.

Table 63. Attributes of the <priority> element

Name	Type	Use	Description
`high`	string	required	ID of the prioritized road
`low`	string	required	ID of the road with lower priority

Rules

The following rules apply to priorities of roads within a junction:

- asam.net:xodr:1.7.0:junctions.priority.no_signals: `<priority>` elements should only be used if there are no signals defined.
- asam.net:xodr:1.8.0:junctions.priority.high_and_low_attr: `<priority>` elements shall be defined with a pair of one @high and one @low attribute.

Related topics

- Section 12.1, "Introduction to junctions"

12.4.2 Direction of connecting roads

Connecting roads inside a junction may have different directions. For ease of use, the road reference line of the connecting roads should be placed in driving direction if the driving direction is unique.

The @contactPoint attribute inside the `<connection>` element is used to specify the direction of a connecting road.

Rules

The following rules apply to the direction of connecting roads:

- asam.net:xodr:1.7.0:junctions.connection.start_along_linkage: The value `start` shall be used to indicate that the connecting road runs along the linkage indicated in the `<laneLink>` element.
- asam.net:xodr:1.7.0:junctions.connection.end_opposite_linkage: The value `end` shall be used to indicate that the connecting road runs along the opposite direction of the linkage indicated in the `<laneLink>` element

Related topics

- Section 10.4, "Road type"